LAB 02 REPORT TEMPLATE

Advanced API Testing & Automation with Postman (Medusa.js Staging)

Course Name & Code:	Advanced Testing & QA Automation
Team Name / ID:	Team03
Jira / Trello Board Link:	https://vanlangsaigon.atlassian.net/jira/software/projects/ATM/boards/37/backlog?selectedIssue=ATM-34
GitHub Repository URL:	https://github.com/Swam-Htet-Paing/ATQAA_Team03_Medusa_Automation

SECTION I: TEAM ROSTER & PEER ASSESSMENT MATRIX

Complete the allocation table below. All team members must agree on contribution percentages and sign prior to submission.

No.	Student Name	Student ID	Assigned Responsibilities	Contrib.	Sign
1	Swam Htet Paing	2500IT10007	Ran collection tests with newman, exported environment and collection json, made the report	35%	Swam
2	Min Kaung San	2500IT10006	Set up Postman environment, collection	32.5%	Min
3	Thu Ta Lin	2500IT10005	Ran tests and configured endpoints	32.5%	Thu

SECTION II: WORKSPACE & ENVIRONMENT CONFIGURATION

Configure Postman Environment variables to ensure zero hardcoded values in API requests.

Variable Name	Initial Value	Current Value	Variable Scope & Description
baseUrl	https://atqaa-medusa-store.onrender.com/store	https://atqaa-medusa-store.onrender.com/store	Global/Env: Storefront REST API base URL
adminBaseUrl	https://atqaa-medusa-store.onrender.com	https://atqaa-medusa-store.onrender.com	Admin Global/Env: REST API base URL
apiToken	blank	eyJhbGciOiJIUzI1NiIsInR5cCI6Ikp...	Env: JWT Bearer Toke

			n for Admin authorization
variantId	blank	variant_01M4CX92CRHTENEF0SNNWNR6EF	Env: Dynamic Product Variant ID for cart line items
cartId	blank	cart_01M75CN1679TQTGS1E6YAE08K4	Env: Dynamic Cart ID maintained across the checkout session

orderId	blank	order_01M75A1N8PSR5TE7DQV2ADJVKZ	Env: Dynamic Order ID extracted after completing order checkout
adminPassword	vlscStudent123456	vlscStudent123456	Env: Admin user account authentication password
publishable_api_key	pk_c6c27bfcd5b891a5cc2f4aec1e85a48a414a1368f4...	pk_c6c27bfcd5b891a5cc2f4aec1e85a48a414a1368f4...	Global/Env: Medusa Publishable API

			Key for storefront requests
shippingOptionId	blank	so_01M4YHKCKQZ4VXFJE7GM4GYNSZ	Env: Shipping Option ID selected for order fulfillment
region_id	blank	reg_01M4BAM8S1CM467KDCWWDKVDWW	Env: Storefront Region ID determining currency and region

			settings
paymentCollectionId	blank	pay_col_01M75A1HYHS7XS4HS1G64XW25G	Env: Payment Collection ID created to handle cart payment workflow
provider_id	pp_system_default	pp_system_default	Global/Env: Payment Provider ID used to initialize payment

			session
lineItemId	blank	cali_01M75CN87BPGFE8W3JXJ471CXS	Env: Line Item ID added to the current active cart
draftOrderId	blank	blank	Env: Dynamic Draft Order ID reserved for admin order creation

SECTION III: DYNAMIC API CHAINING WORKFLOW (5 REQUESTS)

Detail the sequence of 5 E2E chained API requests simulating a complete user purchasing journey.

Seq	HTTP Method & Endpoint	Input Variables	Extracted Dynamic Output	Verification Status
1	GET {{baseUrl}}/products	baseUrl, publishable-api-key	variantId (extracted dynamically)	Tier 1-3 + Ajv JSON Schema validation
2	GET {{baseUrl}}/product-variants	baseUrl	variantId (backup set)	Status 200 OK + variantId environment set
3	POST {{baseUrl}}/carts	baseUrl, publishable-api-key	cartId (extracted dynamically)	Status 200/201 + cartId environment set
4	GET {{baseUrl}}/carts/{{cartId}}	baseUrl, cartId	-	[Contract Validation] Ajv JSON Schema for Cart
5	POST {{baseUrl}}/carts/{{cartId}}/line-items	baseUrl, cartId, variantId	-	Tier 1-4 + DDT item_qty CSV validation
6	POST {{baseUrl}}/carts/{{cartId}}/shipping-methods	baseUrl, cartId, shippingOptionId	-	Status 200/201 shipping method selection
7	POST {{baseUrl}}/payment-collections	baseUrl, cartId, publishable_api_key	paymentCollectionId	Extract paymentCollectionId for payment session
8	POST {{baseUrl}}/payment-collections/{{paymentCollectionId}}/payment-sessions	baseUrl, paymentCollectionId, provider_id	-	Initialize payment session with default provider
9	POST {{baseUrl}}/carts/{{cartId}}/complete	baseUrl, cartId	orderId (extracted dynamically)	Status 200 OK + Order extraction validation
10	POST {{adminBaseUrl}}/auth/user/emailpass	adminBaseUrl, adminPassword	apiToken (Bearer Token)	Status 200 OK + Admin JWT Token set

SECTION IV: MULTI-TIER JAVASCRIPT ASSERTIONS & CODE SNIPPETS

Paste key JavaScript assertion snippets implemented in Postman Tests tab for all 5 requests.

1. Multi-Tier Global & Request Assertions

```
// Global Multi-Tier Assertions (Collection Level)
pm.test("[Tier 1] HTTP Status is successful (200 OK or 201 Created)", function () {
    pm.expect(pm.response.code).to.be.oneOf([200, 201]);
});

pm.test("[Tier 2] Response Time complies with SLA (< 2000ms)", function () {
    pm.expect(pm.response.responseTime).to.be.below(2000);
});

pm.test("[Tier 3] Content-Type header is application/json", function () {
    pm.expect(pm.response.headers.get("Content-Type")).to.include("application/json");
});
```

2. Ajv JSON Schema Contract Validation Snippets

```
// Contract Validation: GET /products Schema
const Ajv = require('ajv');
const ajv = new Ajv();
const productCatalogSchema = {
            "$schema": "http://json-schema.org/draft-07/schema#",
            "type": "object",
            "required": ["products"],
            "properties": {
            "products": {
            "type": "array",
            "items": {
            "type": "object",
            "required": ["id", "title", "handle", "variants"]
            }
            }
            }
};
const validateProducts = ajv.compile(productCatalogSchema);
pm.test("[Contract Validation] GET /products adheres to Ajv JSON Schema", function () {
    pm.expect(validateProducts(pm.response.json())).to.be.true;
});

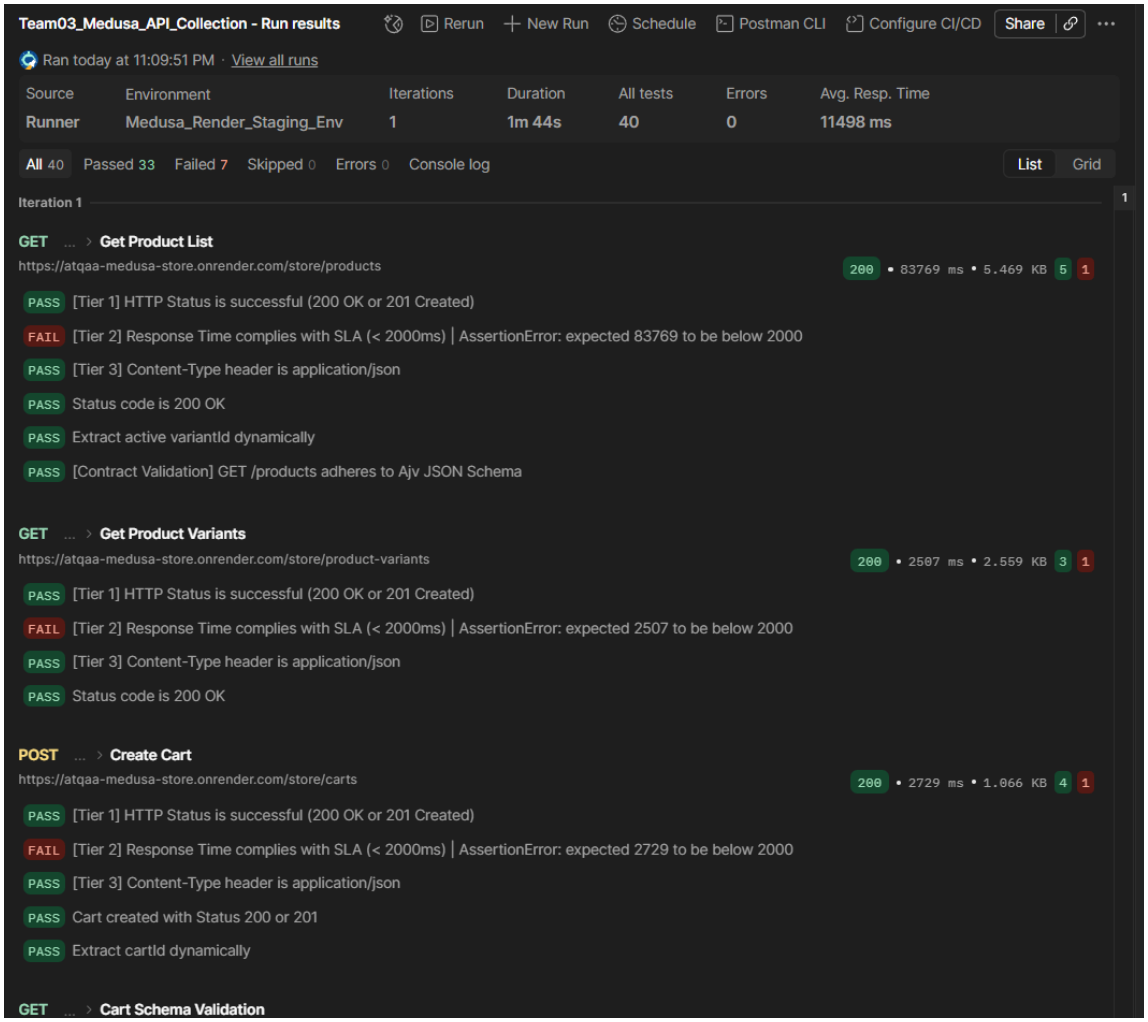
// Contract Validation: GET /carts/{{cartId}} Schema
const cartResponseSchema = {
```

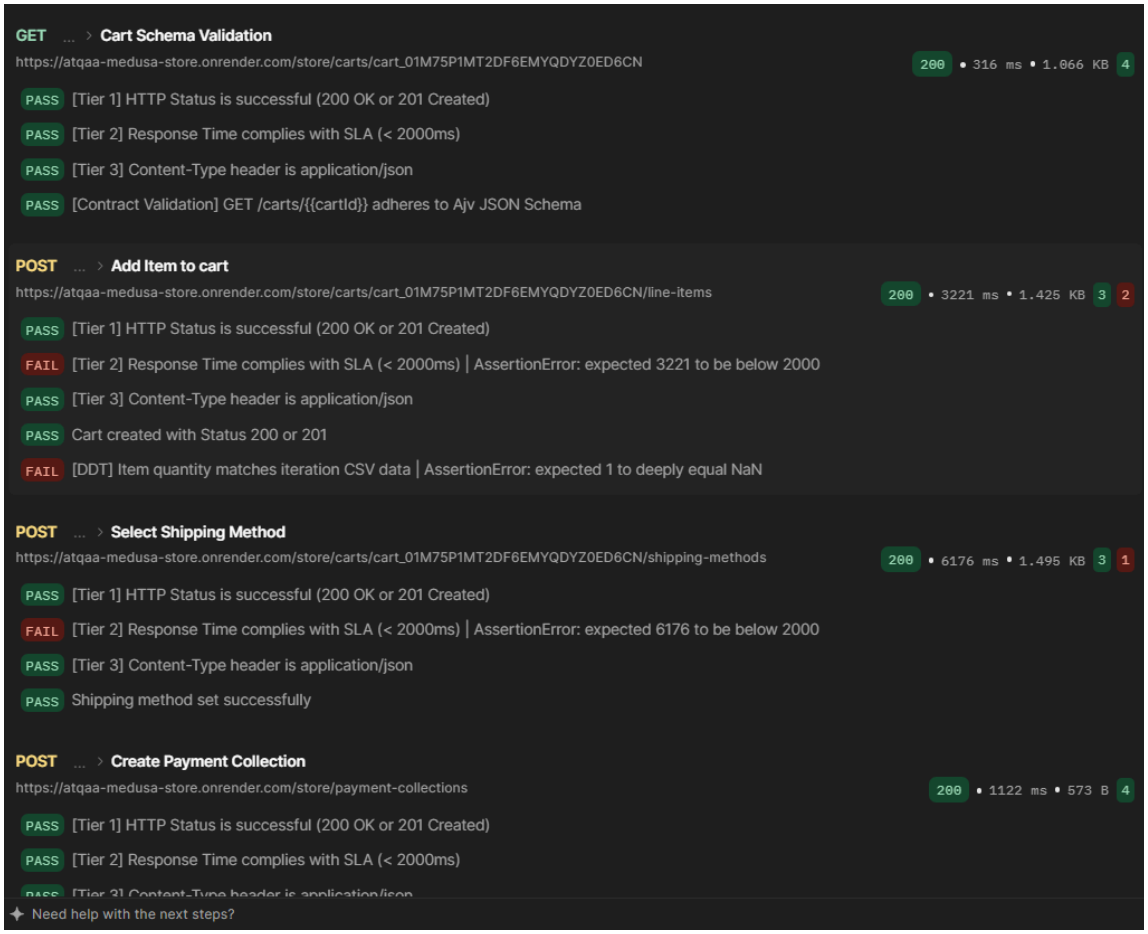
```
            "type": "object",
            "required": ["cart"],
            "properties": {
            "cart": {
            "type": "object",
            "required": ["id", "items", "region_id", "currency_code"]
            }
            }
};
pm.test("[Contract Validation] GET /carts/{{cartId}} adheres to Ajv JSON Schema", function () {
    pm.response.to.have.jsonSchema(cartResponseSchema);
});
```

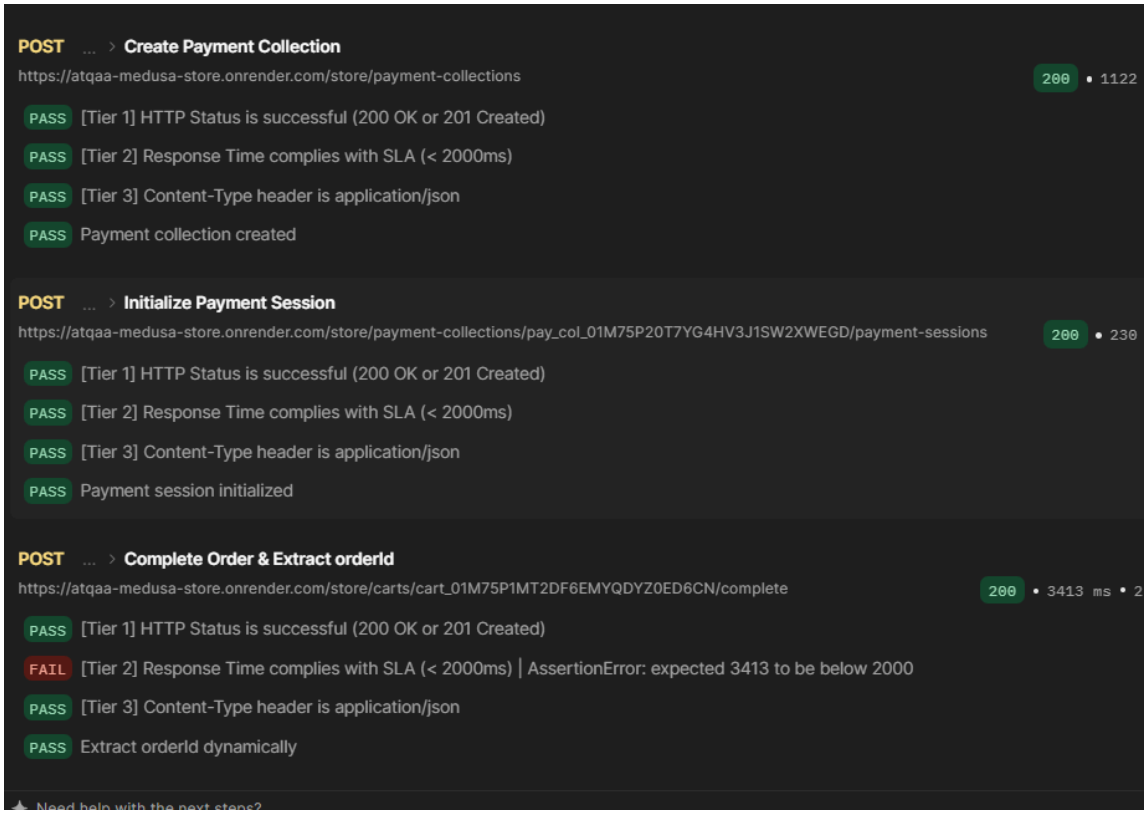
SECTION V: COLLECTION RUNNER & NEWMAN CLI EXECUTION EVIDENCE

Attach evidence screenshots of Postman Collection Runner passing 100% and Newman CLI terminal output.

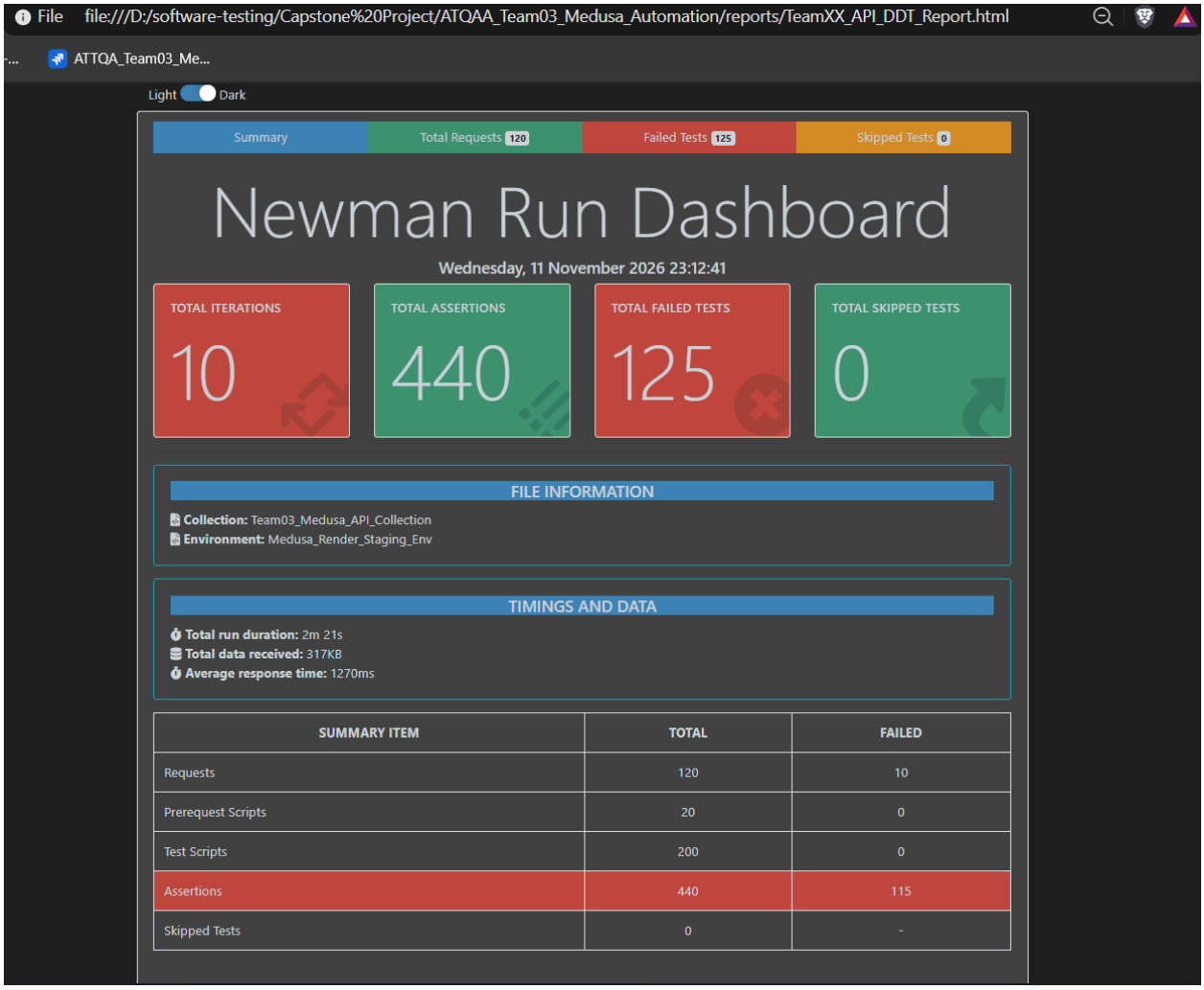
SCREENSHOT 1: Postman Collection Runner — 100% PASSED Results


Team03_Medusa_API_Collection - Run results
Rerun
New Run
Schedule
Postman CLI
Configure CI/CD
Share
Ran today at 11:09:51 PM · View all runs
Source
Runner
Environment
Medusa_Render_Staging_Env
Iterations
1
Duration
1m 44s
All tests
40
Errors
0
Avg. Resp. Time
11498 ms
All 40
Passed 33
Failed 7
Skipped 0
Errors 0
Console log
List
Grid
Iteration 1
1
GET ... › Get Product List
https://atqaa-medusa-store.onrender.com/store/products
200 • 83769 ms • 5.469 KB 5 1
PASS [Tier 1] HTTP Status is successful (200 OK or 201 Created)
FAIL [Tier 2] Response Time complies with SLA (< 2000ms) | AssertionError: expected 83769 to be below 2000
PASS [Tier 3] Content-Type header is application/json
PASS Status code is 200 OK
PASS Extract active variantId dynamically
PASS [Contract Validation] GET /products adheres to Ajv JSON Schema
GET ... › Get Product Variants
https://atqaa-medusa-store.onrender.com/store/product-variants
200 • 2507 ms • 2.559 KB 3 1
PASS [Tier 1] HTTP Status is successful (200 OK or 201 Created)
FAIL [Tier 2] Response Time complies with SLA (< 2000ms) | AssertionError: expected 2507 to be below 2000
PASS [Tier 3] Content-Type header is application/json
PASS Status code is 200 OK
POST ... › Create Cart
https://atqaa-medusa-store.onrender.com/store/carts
200 • 2729 ms • 1.066 KB 4 1
PASS [Tier 1] HTTP Status is successful (200 OK or 201 Created)
FAIL [Tier 2] Response Time complies with SLA (< 2000ms) | AssertionError: expected 2729 to be below 2000
PASS [Tier 3] Content-Type header is application/json
PASS Cart created with Status 200 or 201
PASS Extract cartId dynamically
GET ... › Cart Schema Validation


GET ... > Cart Schema Validation
https://atqaa-medusa-store.onrender.com/store/carts/cart_01M75P1MT2DF6EMYQDYZ0ED6CN
200 • 316 ms • 1.066 KB 4
PASS [Tier 1] HTTP Status is successful (200 OK or 201 Created)
PASS [Tier 2] Response Time complies with SLA (< 2000ms)
PASS [Tier 3] Content-Type header is application/json
PASS [Contract Validation] GET /carts/{{cartId}} adheres to Ajv JSON Schema
POST ... > Add Item to cart
https://atqaa-medusa-store.onrender.com/store/carts/cart_01M75P1MT2DF6EMYQDYZ0ED6CN/line-items
200 • 3221 ms • 1.425 KB 3 2
PASS [Tier 1] HTTP Status is successful (200 OK or 201 Created)
FAIL [Tier 2] Response Time complies with SLA (< 2000ms) | AssertionError: expected 3221 to be below 2000
PASS [Tier 3] Content-Type header is application/json
PASS Cart created with Status 200 or 201
FAIL [DDT] Item quantity matches iteration CSV data | AssertionError: expected 1 to deeply equal NaN
POST ... > Select Shipping Method
https://atqaa-medusa-store.onrender.com/store/carts/cart_01M75P1MT2DF6EMYQDYZ0ED6CN/shipping-methods
200 • 6176 ms • 1.495 KB 3 1
PASS [Tier 1] HTTP Status is successful (200 OK or 201 Created)
FAIL [Tier 2] Response Time complies with SLA (< 2000ms) | AssertionError: expected 6176 to be below 2000
PASS [Tier 3] Content-Type header is application/json
PASS Shipping method set successfully
POST ... > Create Payment Collection
https://atqaa-medusa-store.onrender.com/store/payment-collections
200 • 1122 ms • 573 B 4
PASS [Tier 1] HTTP Status is successful (200 OK or 201 Created)
PASS [Tier 2] Response Time complies with SLA (< 2000ms)
Need help with the next steps?



SCREENSHOT 2: Newman CLI Execution in Terminal / HTML Report Output


File file:///D:/software-testing/Capstone%20Project/ATQAA_Team03_Medusa_Automation/reports/TeamXX_API_DDT_Report.html
ATTQA_Team03_Me...
Light Dark
Summary
Total Requests 120
Failed Tests 125
Skipped Tests 0
Newman Run Dashboard
Wednesday, 11 November 2026 23:12:41
TOTAL ITERATIONS
10
TOTAL ASSERTIONS
440
TOTAL FAILED TESTS
125
TOTAL SKIPPED TESTS
0
FILE INFORMATION
Collection: Team03_Medusa_API_Collection
Environment: Medusa_Render_Staging_Env
TIMINGS AND DATA
Total run duration: 2m 21s
Total data received: 317KB
Average response time: 1270ms
SUMMARY ITEM
TOTAL
FAILED
Requests
120
10
Prerequest Scripts
20
0
Test Scripts
200
0
Assertions
440
115
Skipped Tests
0
-